

Team 39 AI Strategy

Use Case Exploration Worksheet

Objective: To explore and identify viable data-driven sustainability use cases for our project.

Instructions: This worksheet is divided into two main parts. Part 1 focuses on brainstorming and detailing potential use cases, while Part 2 is optional and delves into a specific use case that seems most prevalent. Please document your findings and thoughts in a word document, ensuring clarity and thoroughness in your exploration.

Part 1: Broad Exploration

1. Scenario Identification

- Come up with various scenarios within the domain of data-driven sustainability. Examples might include travel emissions, energy consumption analysis, etc. Describe each scenario briefly, highlighting its relevance to sustainability.

2. Main Use Case Selection

- From the scenarios listed above, select the one you find most compelling and suitable, considering potential data sources and the impact of analysis. Break down this use case into the following:

- **Activity Description:** Describe the activities involved.

- **Data Collection:** Specify the data to be collected directly from users and the datasets required for AI analysis.

- **AI Analysis:** Discuss how the AI will process this data, comparing it to existing datasets.

- **Stakeholder Benefit:** Explain the potential benefits of this analysis for stakeholders.

3. Advanced Research

- Attempt to identify actual datasets that could support your chosen use case. Consider whether surveys could be a viable method for gathering additional data.

Part 2: Optional - Detailed Exploration of a Prevalent Use Case

If your chosen use case in Part 1 is different from the travel to office scenario, which has been identified as a prevalent option, please conduct a similar detailed investigation focusing specifically on this use case.

- **Feasibility Study:** Carry out an in-depth feasibility study for the travel to office use case. Apply the same breakdown as in Part 1 to this scenario.

- **Submission:** Compile your findings and analysis into a comprehensive document. Overlap with other team members' work is acceptable, as this will aid in our collective understanding and approach to the chosen problem.

My idea from the meeting meeting

3) How much carbon emissions are made from travelling to work - Can also ask people in person - in forms, can populate the tables that the ai is trained on with this info

- Input data for the AI - method of transport and time taken

- Can use a maps api to get the distance travelled

- Makes the individual accountable for their carbon emissions

- Working from home vs working in office - in stats

- Find the individual stats for if people drive - ask them the model of their cars - pull the carbon efficiency of their car - Mati found a good dataset

- Addition - if the company has an overall carbon goal for a month,

The webapp can help them meet those goals.

- A google form that will contain a few questions, like how they got to uni, how long it took them.

- Future Projections

- Potential Idea - based on user input - put it on a bell shaped curve - if they are in a smaller percentile, then they should be issued with warnings/comments.

Written By Andreas Sergiou

1) Ideas for Use cases:

- **Personal Employee Carbon Tracker (PECT)** - How much carbon emissions are produced from travelling to work. By analysing how everyone gets to work, in what way and how long they travel for, we can estimate their carbon emission production and show potential monthly estimates if they carry on doing this journey, or what it could look like if they took other modes of transport, or different routes.

- **Smart Building Management for Energy Efficiency** - How energy efficient a business building is - can use the data from the Eviden building in Nottingham - Building have technologies like, IoT sensors and intelligent HVAC systems, which would collect data on how many people there are in the building, the temperature of the whole building, and energy use. If we were to take these into account and produce an AI based on these factors and produce graphs that will allow businesses to adjust settings in their business buildings in real-time for maximum efficiency and reducing energy waste, all contributing to the sustainability goals of the company.

- **Renewable Energy Integration and Optimization** - Utility companies and energy managers can use data analytics to better integrate renewable energy sources, such as wind and solar, into the power grid. By analysing weather patterns, energy demand, and production capacity, they can optimise the mix of renewable and traditional energy sources to reduce reliance on fossil fuels and lower carbon emissions (This has been done by the team and they said there is not a lot of data to find on it)

2) Main Use Case Selection

- Activity Description:

- PECT is designed to measure and analyse the carbon emissions produced by employees commuting to work. This would assess various modes of transport - car, train, bus, cycling, walking, etc - and how long the person is travelling for to calculate their carbon footprint. PECT aims to provide an insight into better, more-sustainable commuting practices by also suggesting comparisons and potential improvements based on other modes of transport or routes.

- Data Collection:

- **Directly from Users** - Employees would input their daily commuting details, including the mode of transport, distance travelled and time travelling.

- For people who drove to work, more information will have to be required like the vehicle type, fuel type and average fuel consumption

- Datasets Required for AI Analysis:

- **Transport Emission Factors** - A dataset containing average emission factors for various modes of transport, taking into account fuel types and vehicle efficiency.

- **Geospatial Data** - Maps and routing information to calculate distances and possible alternative routes - can use some sort of API to calculate the distance between two points.

- **Public Transport Schedules and Routes** - Information on public transportation options, including routes, frequency, and the environmental footprint of different services - will be different for trains and buses.

- **Weather and Traffic Patterns** - Historical and real-time data to analyse the impact of these factors on commuting choices and emissions

- AI Analysis - How the AI in PECT would process the collected data

- **Emission estimation** - This will calculate the carbon emissions for each employee's reported commute using emission factors for the specified transport mode and distance. This involves converting all the travel information into CO2 equivalent emissions.

- **Alternatives Analysis** - Compare the user's current information that they have inputted with alternatives. This can be done by using geospatial data to suggest more efficient or lower-emission routes and by using public transport schedules to propose better, realistic public transportation options.

- **Trend Prediction** - Analysing the data the user has inputted, try to identify patterns and predict future emissions based on their current habits. Could help in forecasting the potential environmental impact and with this data, the user can try and stop preventing as much CO2 emissions.

- **Personalised Recommendations** - Could offer tailored suggestions for each employee to reduce their carbon footprint, like switching to greener modes of transport, sharing a car with another employee or adjusting commute times to avoid heavy traffic

- Stakeholder Benefits - For every level in and out of the company

- **Employee** - They will gain an insight into their personal environmental impact and will receive advice on how to reduce their carbon footprint, promoting more sustainable commuting practices.

- **Employer** - They will gain an understanding of the overall carbon footprint associated with their workforce's commute, allowing them to support greener commuting initiatives, bettering the businesses sustainability, and potentially qualify for government environmental incentives and schemes

- **Environmental Organisations** - Benefit from aggregated data on commuting patterns and their environmental impact, supporting broader efforts to reduce transportation-related emissions and advocate for policy changes

3) Advanced Research - Potential Survey Layout and Questions:

- Full survey on the next page

Personal Employee Carbon Tracker (PECT) Commuting Survey

Dear Employee,

As part of our commitment to sustainability and reducing our carbon footprint, we are introducing the Personal Employee Carbon Tracker (PECT). This initiative aims to understand and improve our commuting practices by analysing the carbon emissions associated with travelling to work. Your participation is crucial in helping us make more environmentally friendly commuting choices. Please take a few minutes to complete this survey. Your responses will remain confidential and will be used to calculate your personal commuting carbon footprint and explore sustainable alternatives.

Personal Information

Full Name (Optional): _____

Department/Team: _____

Commuting Habits

Q1) How do you typically commute to work? (Please select all that apply)

- Car (solo) ____
- Carpool ____
- Bus ____
- Train ____
- Bike ____
- Walk ____
- Other (please specify): _____

Q2) If you use a car (solo or carpool), what is the make, model, and year of the vehicle?

Make: _____

Model: _____

Year: _____

Q3) What type of fuel does your vehicle use? (Please select one)

- Gasoline ____
- Diesel ____
- Electric ____
- Hybrid ____
- Other (please specify): _____

Q4) Approximately how many miles do you travel to work one way? _____

Q5) On average, how many days per week do you commute to work? _____

Q6) How long does your typical commute take one way? _____

Commuting Preferences and Flexibility

Q7) Would you be interested in exploring alternative commuting options to reduce your carbon footprint?

- Yes ____
- No ____
- Maybe ____

Q8) If you answered "Yes" or "Maybe" to the previous question, which of the following alternatives would you consider? (Please select all that apply)

- Switching to public transportation ____
- Carpooling with colleagues ____
- Biking ____
- Walking ____
- Telecommuting (if job allows) ____
- Other (please specify): _____

Q9) What are the main factors influencing your choice of commuting method? (Please select all that apply)

- Convenience ____
- Time ____
- Cost ____
- Environmental impact ____
- Availability of alternatives ____
- Other (please specify): _____

Q10) Additional Comments - Please share any additional thoughts or suggestions you have about reducing the carbon footprint of our commuting practices.

Thank you for participating in this survey. Your input is valuable in helping us take meaningful steps toward a more sustainable future.